SANJAY S

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SUMMARY

Intern with hands-on experience specializing in AI-driven models and multimedia processing. Proficient in Python, problem-solving, and innovation. Skilled in project management, strategic planning, and process optimization, driving efficiency and impactful results. Strong analytical and communication skills, with the ability to collaborate, adapt, and contribute effectively to team projects.

EXPERIENCE

Research Intern
June 2024,Chennai, Tamilnadu

Sairam Techno Incubator

- Researched and developed deep learning-based image and video generation models using GANs, diffusion models, and transformers.
- Implemented LLM-driven multimodal understanding systems for text, image, and video data processing.
- Performed fine-tuning, model optimization, and performance benchmarking to enhance generative AI efficiency.
- Conducted quantitative analysis of experimental outputs, ensuring adherence to standard research protocols and reproducibility.
- Collaborated on integrating AI pipelines into prototype applications for real-world use cases

AI Research Intern
June 2024,Chennai, Tamilnadu

Sairam Techno Incubator

- I worked on a client project where I gained experience in how a typical company operates and the steps involved in its processes, along with hands-on experience in developing a pantry manager using diffusion models.

ML Research Intern
March 2026,Chennai, Tamilnadu

Logic Prime

- I worked on fine-tuning an ML model to classify query tickets shared by the customers of Ridewinners.

Hackathons

- Actively participated in, organized, and secured wins in several hackathons.
- I regularly experiment with newly released open-source models locally, especially those from companies like Meta, Hunyuan, and Kimi, and I also create and deploy projects using Hugging Face Spaces.

EDUCATION

M.tech Computer Science and Engineering (5 years - integrated)
May 2027

Sri Sairam Engineering college·Chennai, Tamilnadu

Computer science with Mathematics
May 2022

Sri S Badalchand Sugankavur Chordia Vivekananda Vidyalaya Matriculation Higher Sec School·Chennai, Tamilnadu·May 2022

SKILLS

- Programming Languages: Python, C
- Frameworks & Libraries: TensorFlow, PyTorch, OpenCV, Scikit-learn, Flask
- Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab
- Core Competencies: Object-Oriented Programming (OOP), Problem Solving, Debugging
- Specialized Skills: Deep Learning, Generative AI, Model Fine-Tuning, LLMs, Multimodal Processing, Computer Vision
- Software Development: Application Development, REST API Design, Version Control, Agile Practices
- Operating Systems: Windows

PROJECTS

Ai Virtual Teacher

- Developed an AI-powered communication system that interprets sign language using computer vision and converts it into speech or text in real time. Enabled two-way interaction by generating sign-language responses through multimodal agents to support deaf, mute, and visually impaired users.

AI Photobooth

- Created an AI-powered photo booth using image generation and style transfer models to produce customizable portraits and dynamic visual effects with automated background enhancement.

Agent Orchestration Framework

- Designed an agent orchestration system enabling collaborative AI task execution, combining resource optimization, LLM-based reasoning, and decision automation for workflow management.

Kitchen Pantry Organizer

- Implemented an AI-based pantry management solution leveraging generative models (Gemini, Stable Diffusion) and vector databases to align client products onto a shelf and sync with a product database.

Rooftop Rainwater Harvesting Advisor

- Developed an AI-assisted system that analyzes users' images, satellite, and sensor data to recommend optimal rainwater harvesting setups for urban households, and displays a 3D model of the completed solution for the user.

AI Intent Classifier

- Fine-tuned an ML model to classify query tickets shared by the customers of Ridewinners.